

영상처리 _ BUSINESSCARD

프로젝트(2/2)

21101929 김영환



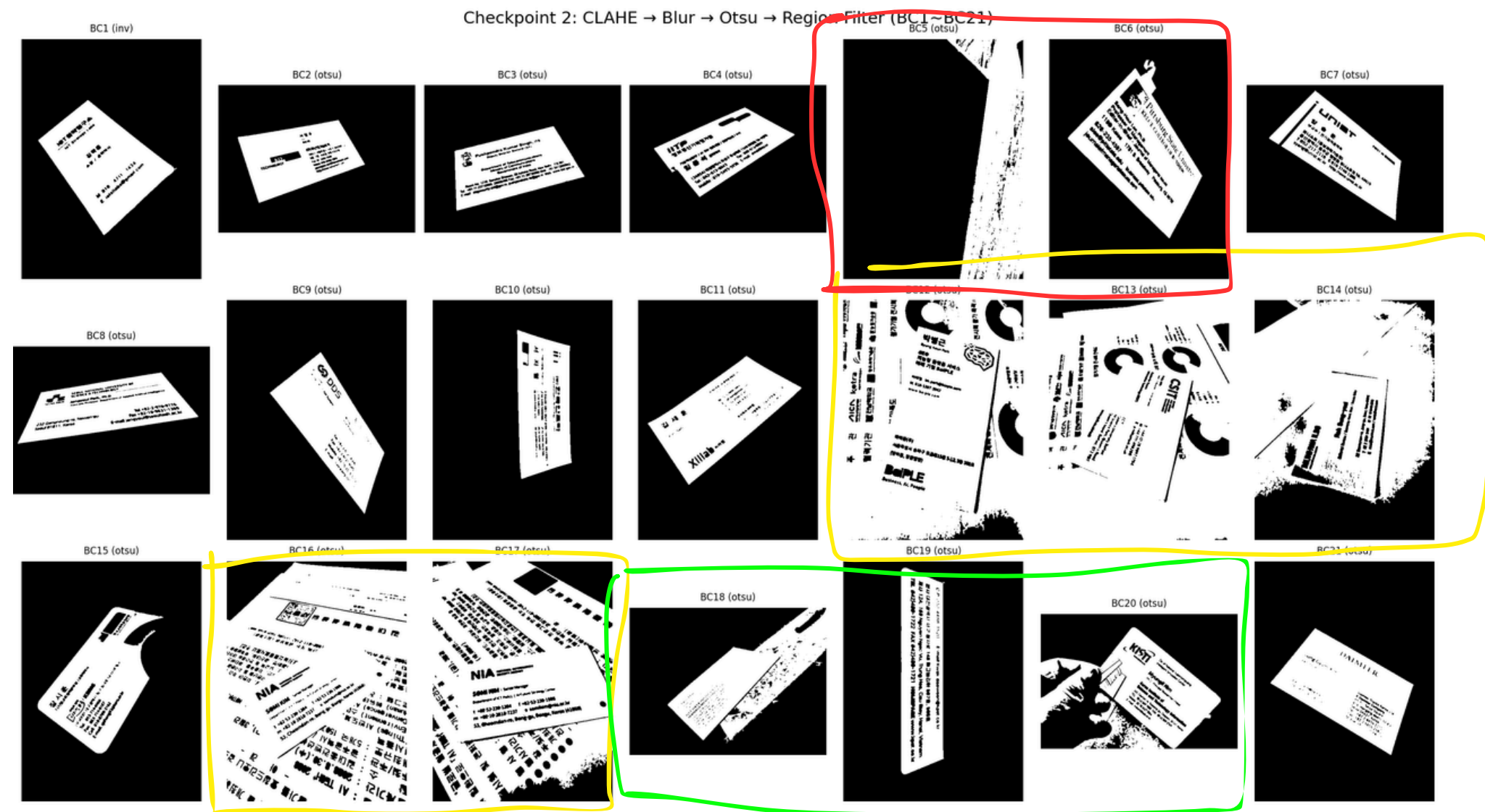
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01 지난 painpoint



(1차 region filter 이후 모습)

1차 단독 pipeline:
otsu 단독->최대 component

한계:
1. 배경이 component
or 배경/사물과 결합
2. 흰배경이면 검출 자체 불가

=> 유형별 type 분리 및 개선

02 typeA/B

기존 1차시: BC1을 위한 하드 코딩

이번 2차시: '명함'이라는 상황에 대한 일반적 상황 가정

(type 구분)

typeB: 기존 배경과 명함이 분명히 구분됨

전체 中 white ratio 60% 이상

→ typeA

CLAHE(32x32) → medianBlur → **Otsu** → 반전 판단 →

(1차시: 16x16)

(1차시: GaussianBlur)

(1차시: WhiteRatio 0.75~0.789 : 하드코딩)

살릴 부분은 살리고 노이즈 연결은 억제하기 위함 2차시: 중앙영역 90% 이상 → 반전(일반화)

fill_holes → region_filter → merge_center_regions → find_corners

(1차시 동일)

(1차시 동일)

명함 내부 디자인으로 분리 가능성

(1차시 동일)

→ 중심영역 해당시 할당(2차시 추가)

02 typeA/B

typeA: (point: 밝은 영역에서 최대한 디테일을 뽑고, '명함'이라는 inductive bias를 최대한 활용하자

gray_scale에서 재시작

↓
clip_black_norm
↓ : 어두운 픽셀(110이하): 0(블랙)으로, 밝은 영역만 → 0~255 선형으로 늘리기

canny
↓ : Sobel → NMS → Hysteresis(strong과 연결)

hough_lines
↓ : k,d = 무한대 각도 불가능 → Hessian 각도 기반 투표

nms_lines
↓ : 각도와 거리 기반 중복 직선 제거 → n개 직선 생존

find_best_quad
↓ : nC4 조합 → 교점 4개 → '명함'스코어

sort_corners
↓ : 행렬 연산용

homography+warp
↓ : typeB와 동일

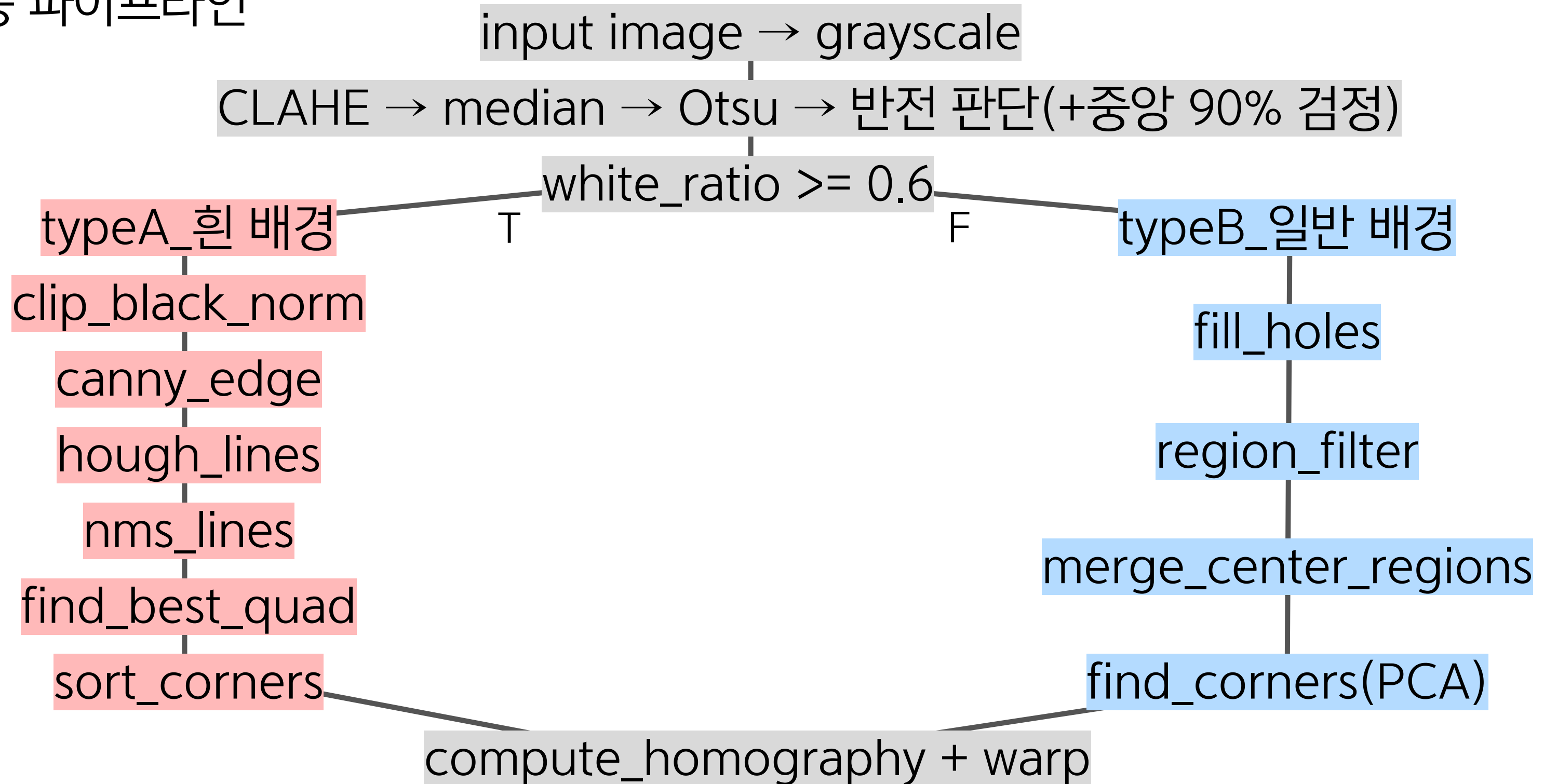
$$\begin{aligned} \text{score} = & \\ & s_area \times s_center \text{ (가중)} \\ & - (\text{asp_penalty} + \text{side_penalty}) \text{ (페널티)} \end{aligned}$$

이때의 inductive bias:

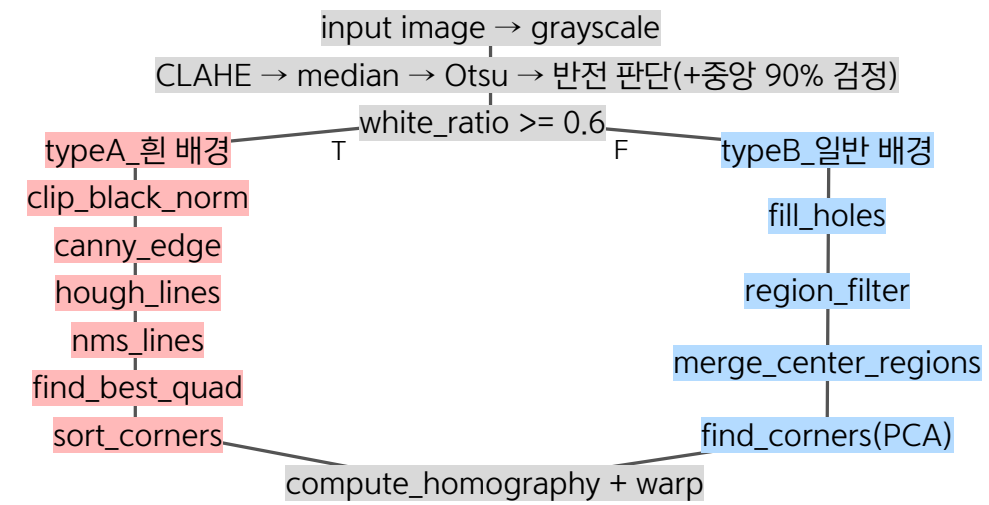
1. 명함 찍는 사람은 명함을 중앙에 두고, 충분히 영역을 고려한다.
2. 명함 종횡비는 약 1.8이다.
3. 명함은 직사각형이므로 대변 길이가 비슷하다.

O2 typeA/B

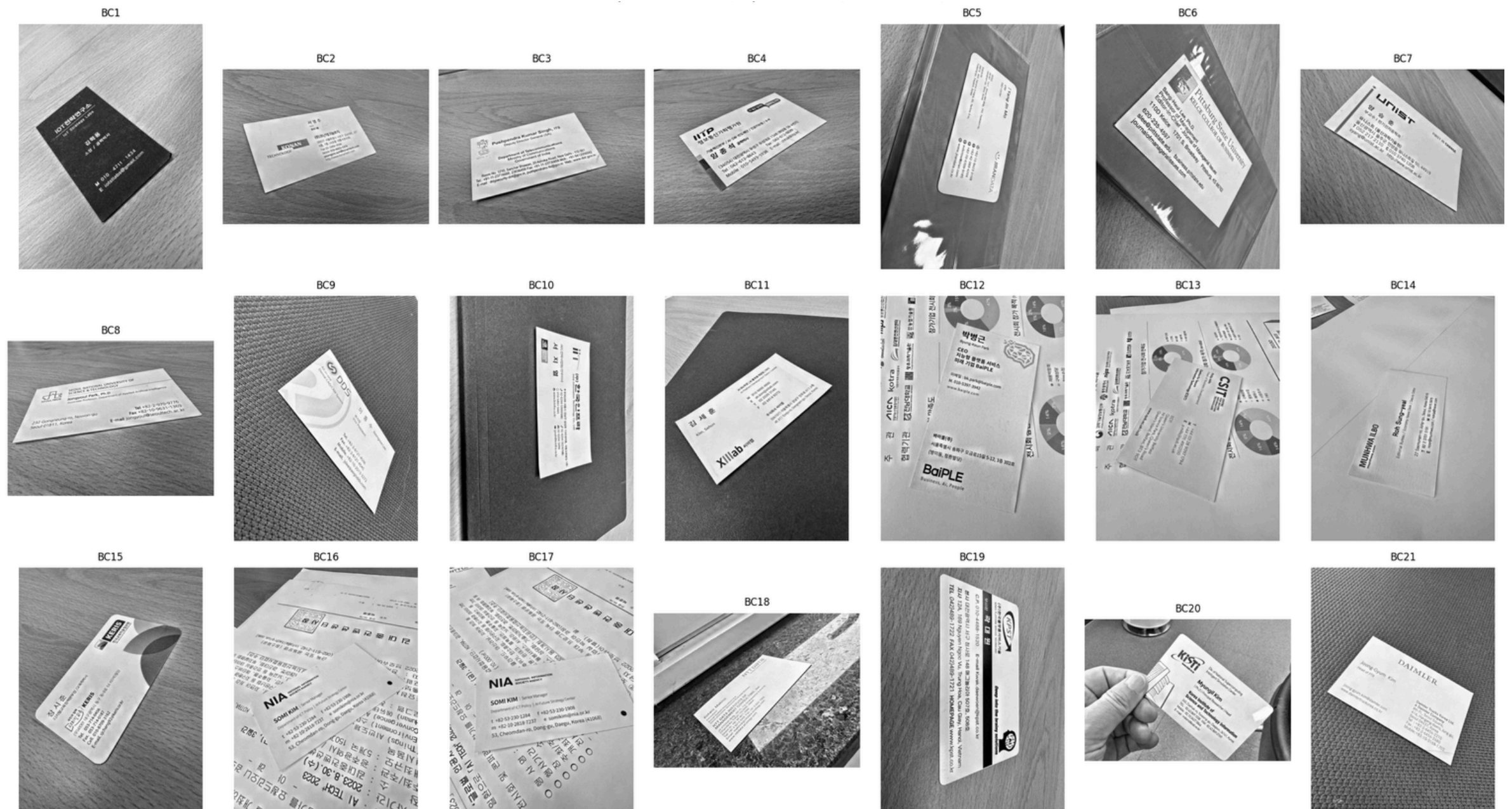
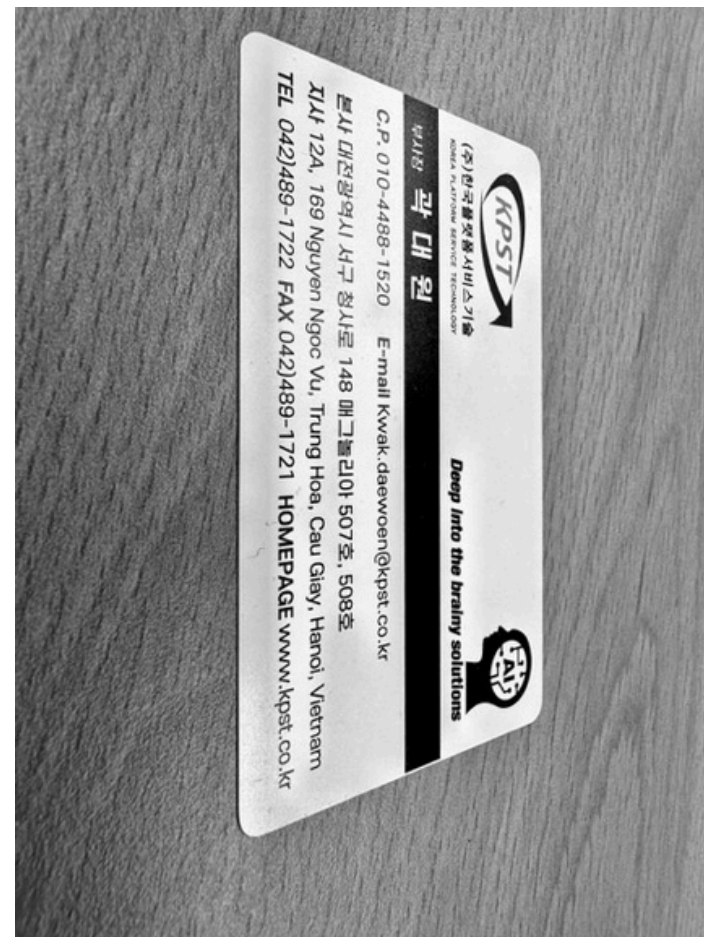
최종 파이프라인



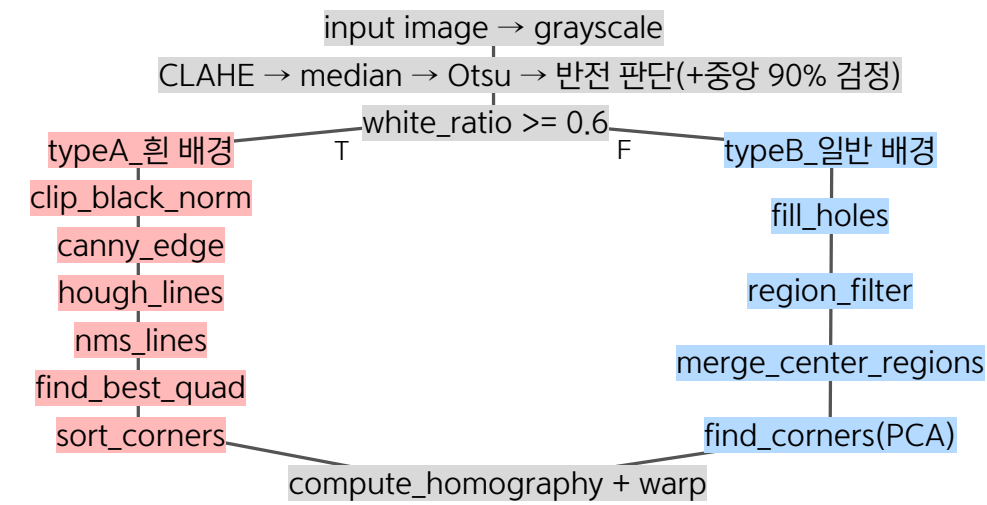
O2 typeA/B



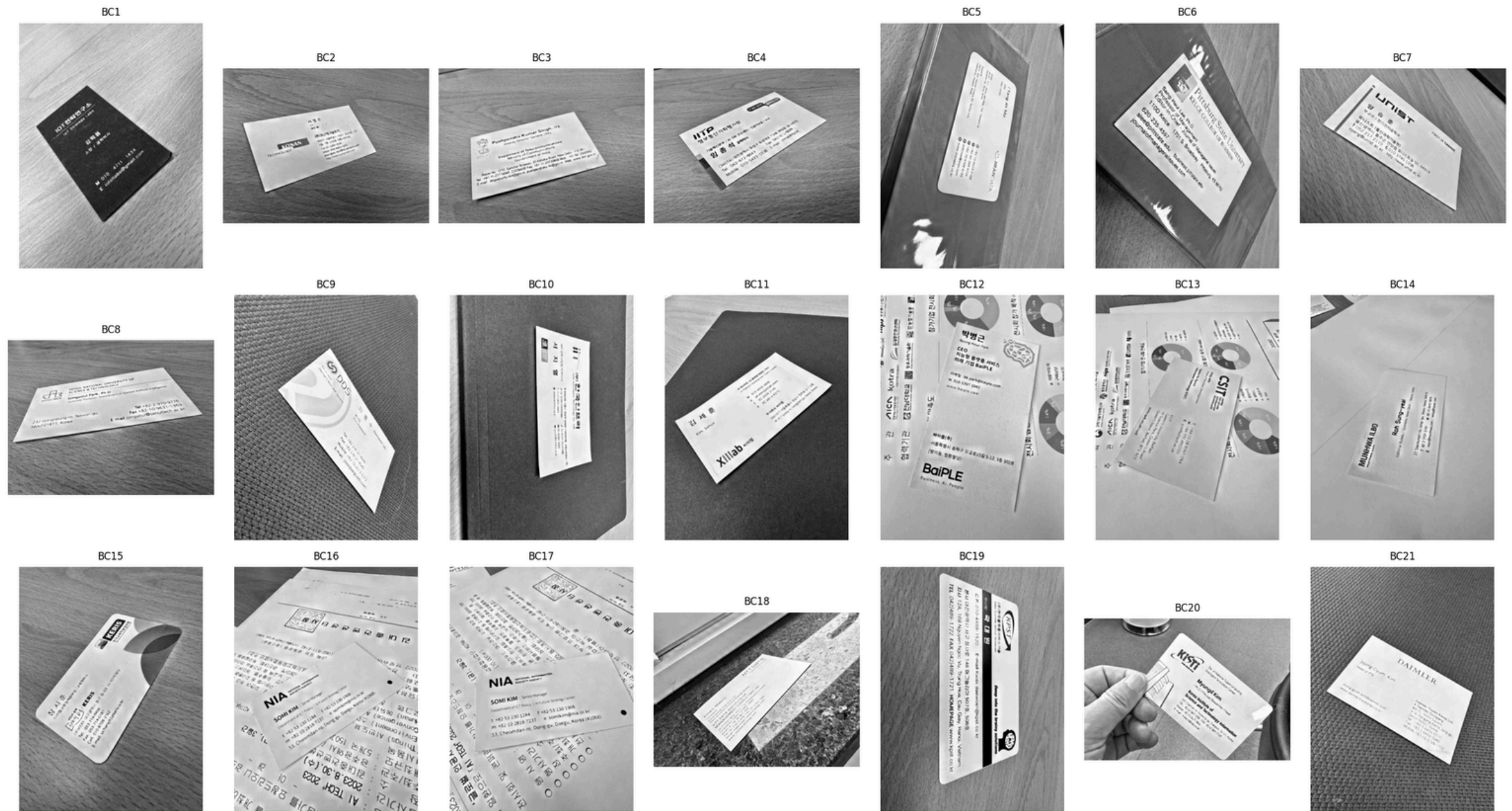
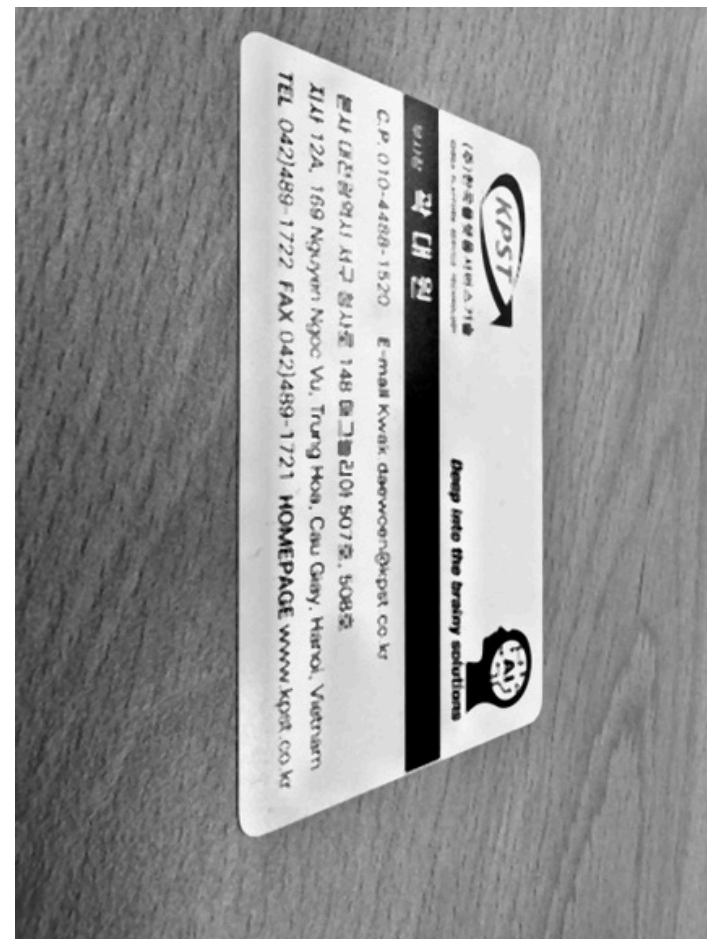
CLAHE (32x32)



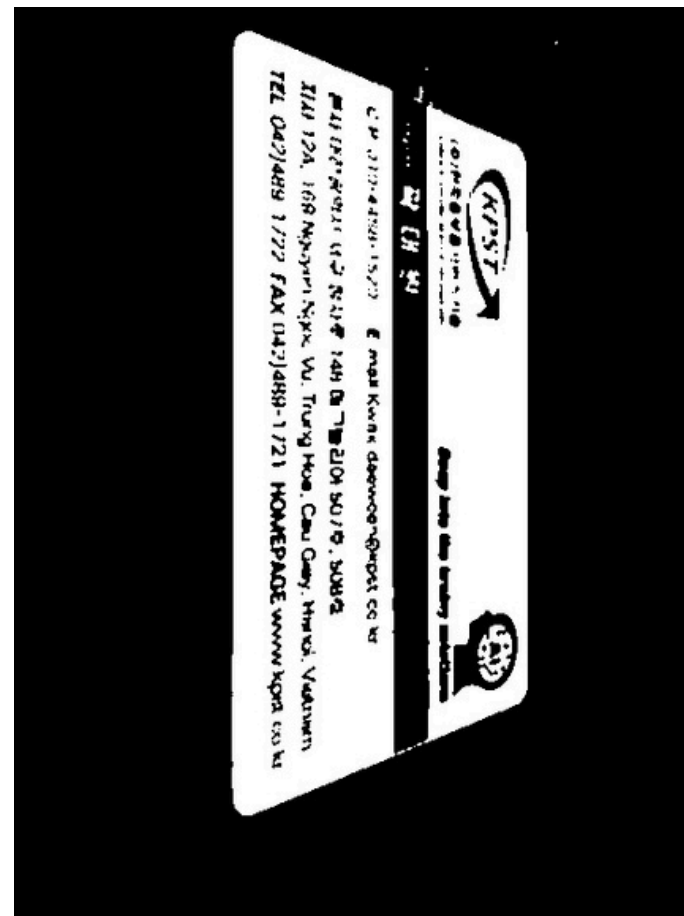
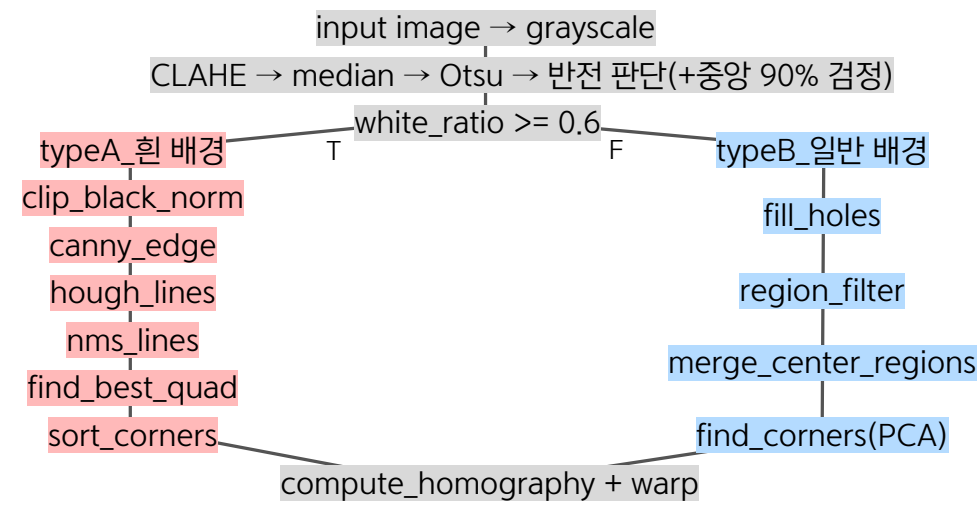
O2 typeA/B



MedianBlur 적용



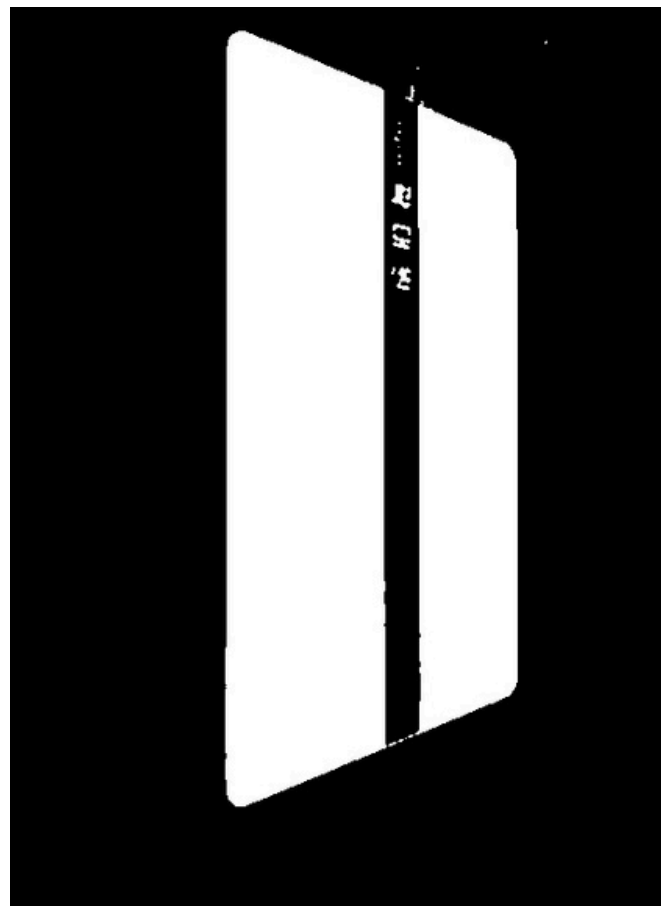
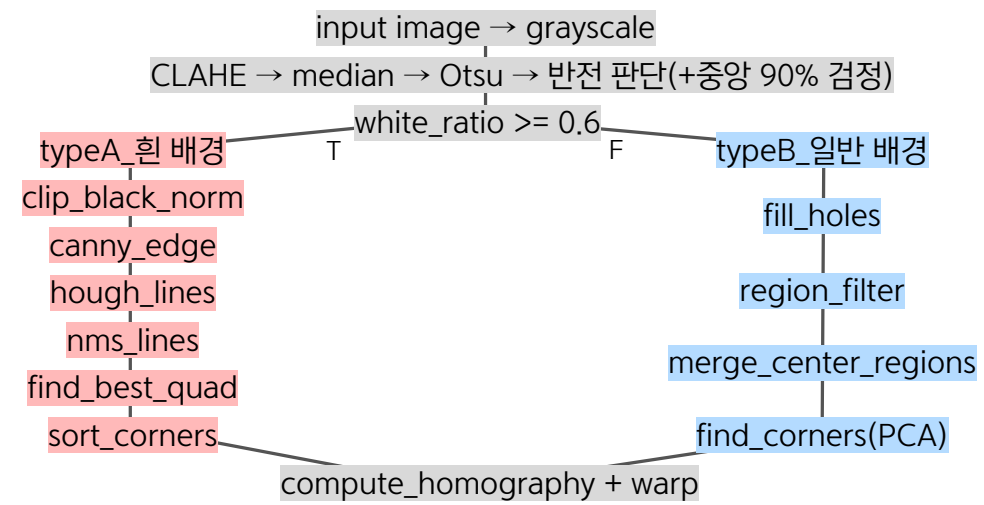
O2 typeA/B



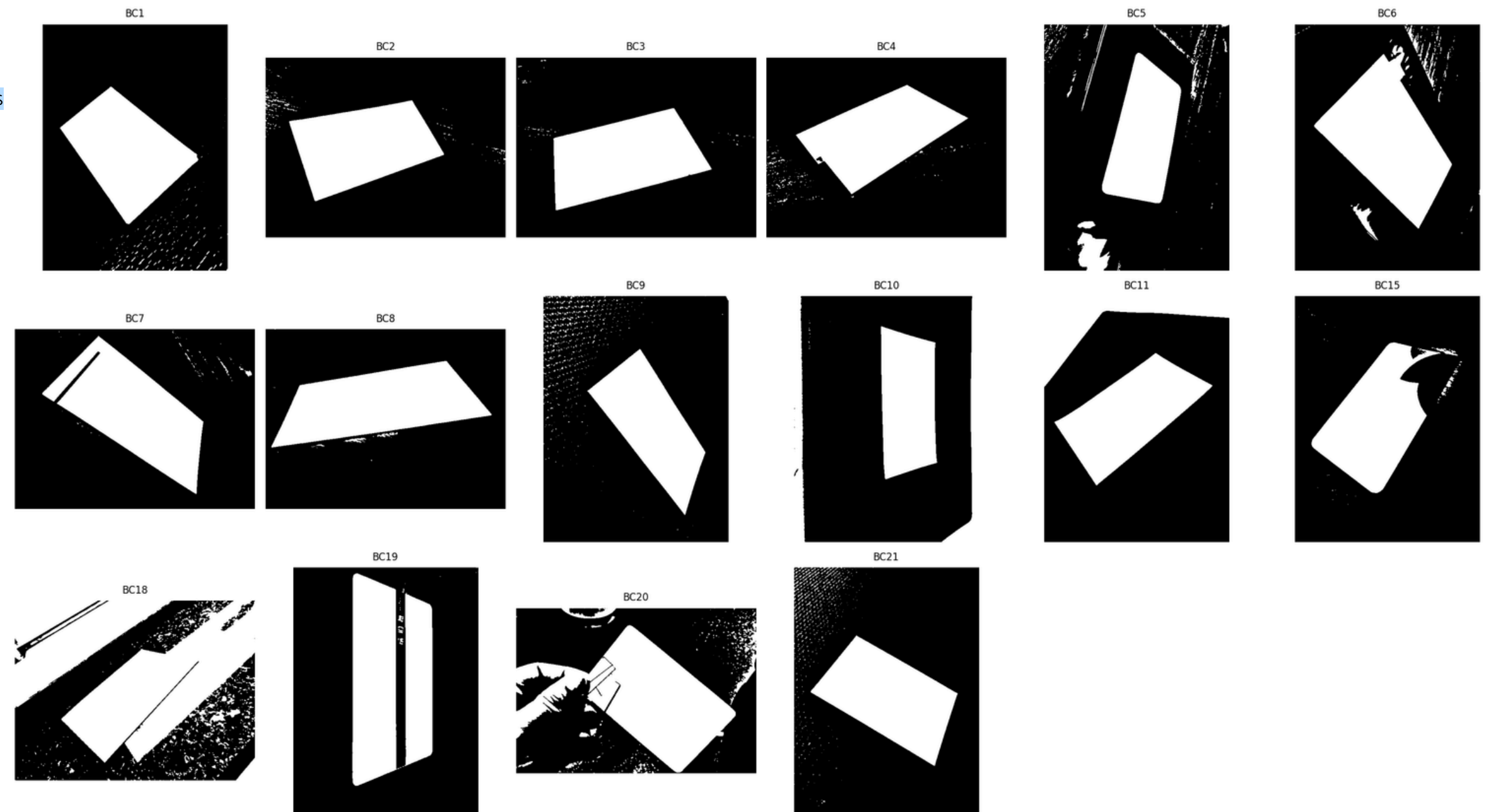
Otsu + 반전 판단(BC1 반전)
 이후 white ratio로 BC12,13,14,16,17 => typeA



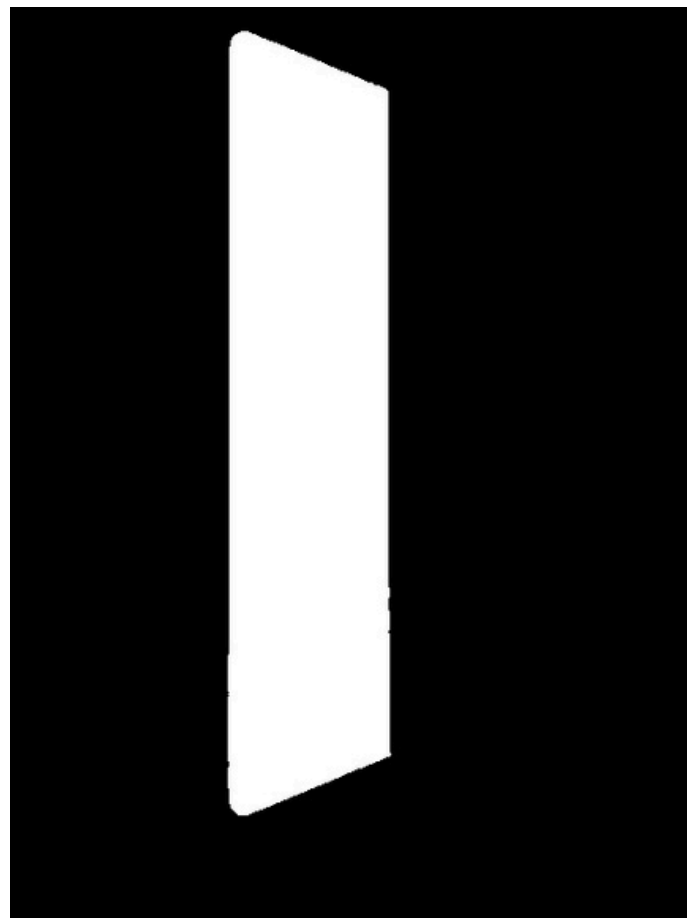
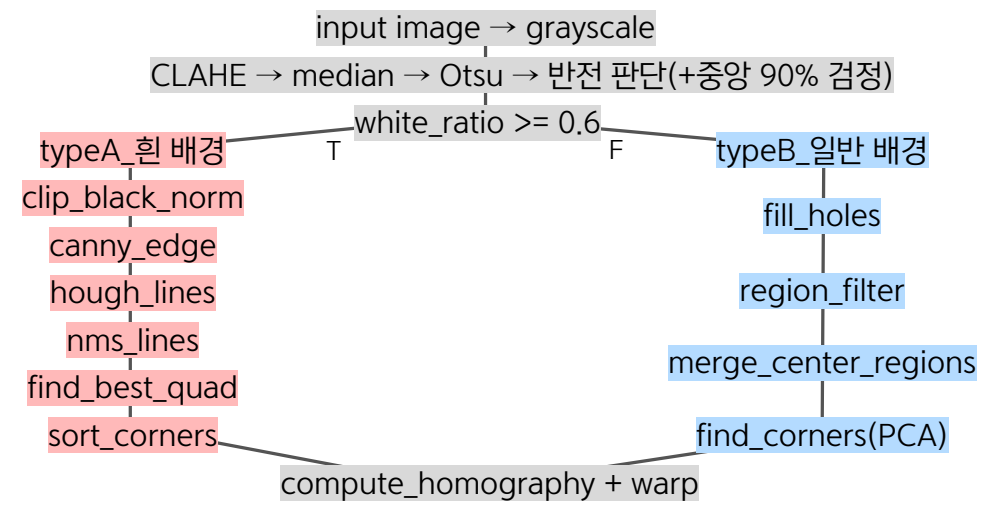
02 typeA/B



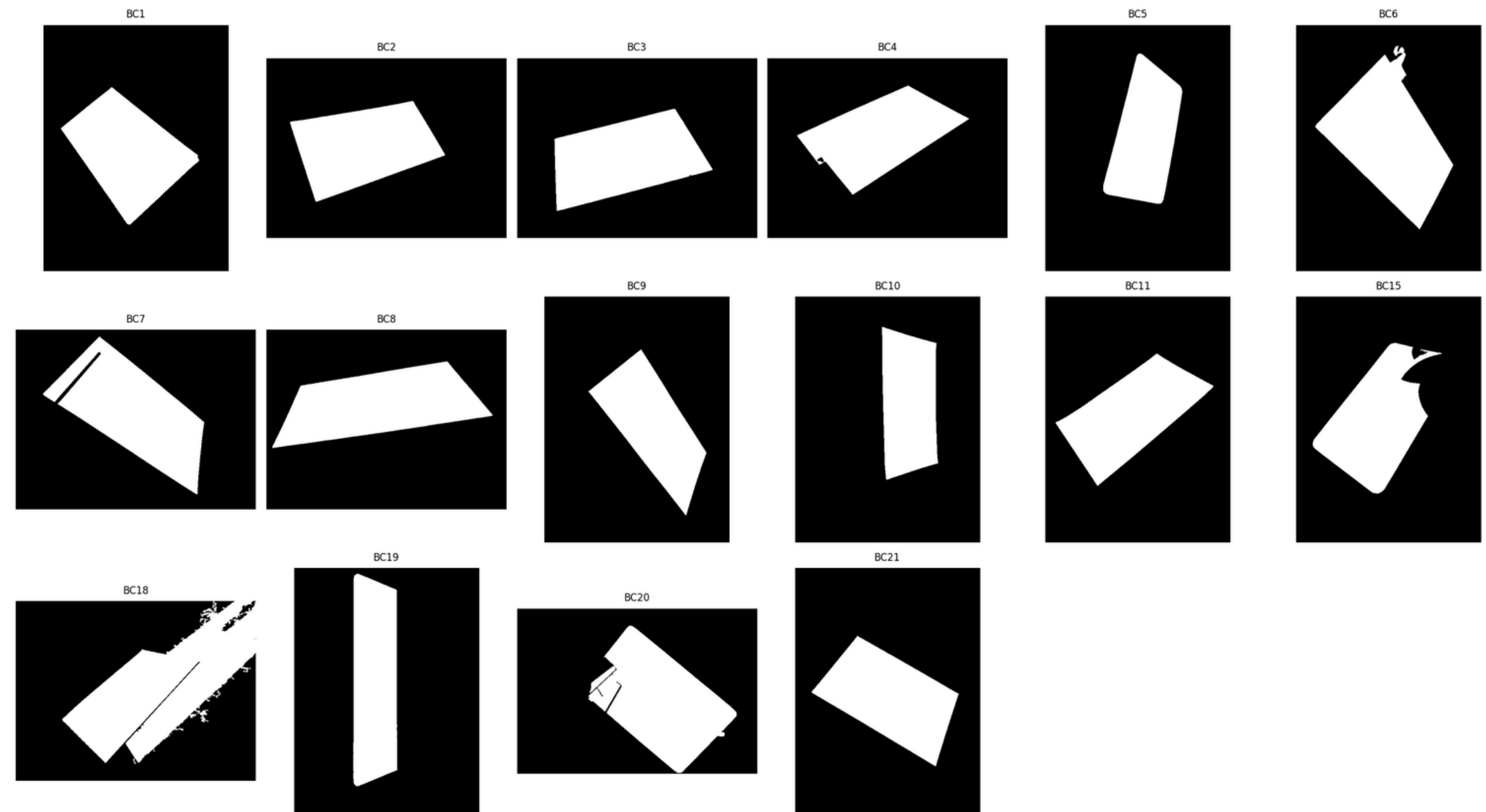
fill_holes: 이미지 테두리에서 출발해 4이웃 BFS로 연결된 검정 픽셀= 배경
배경x → 내부 채우기



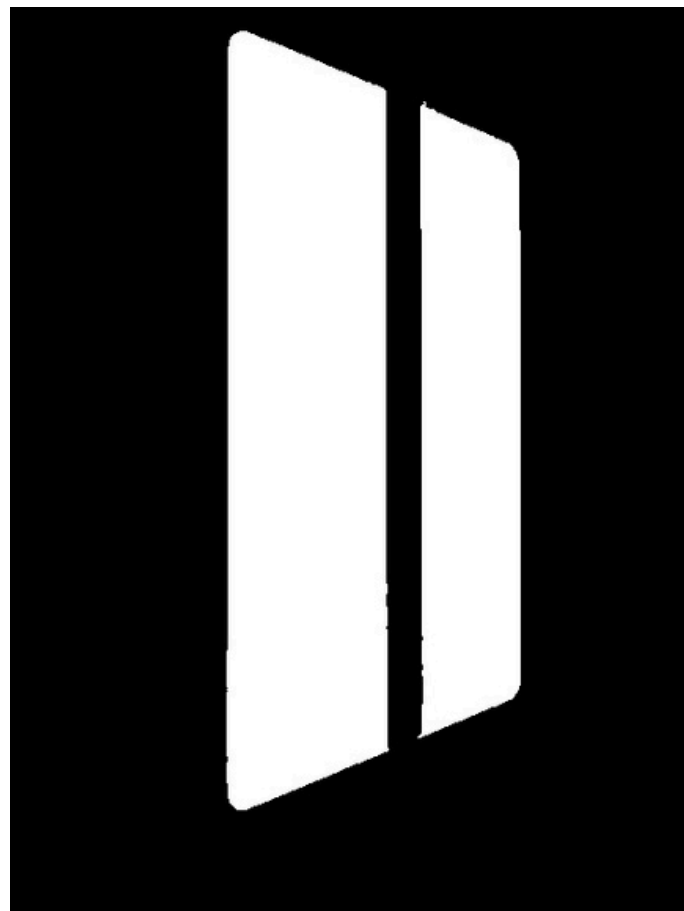
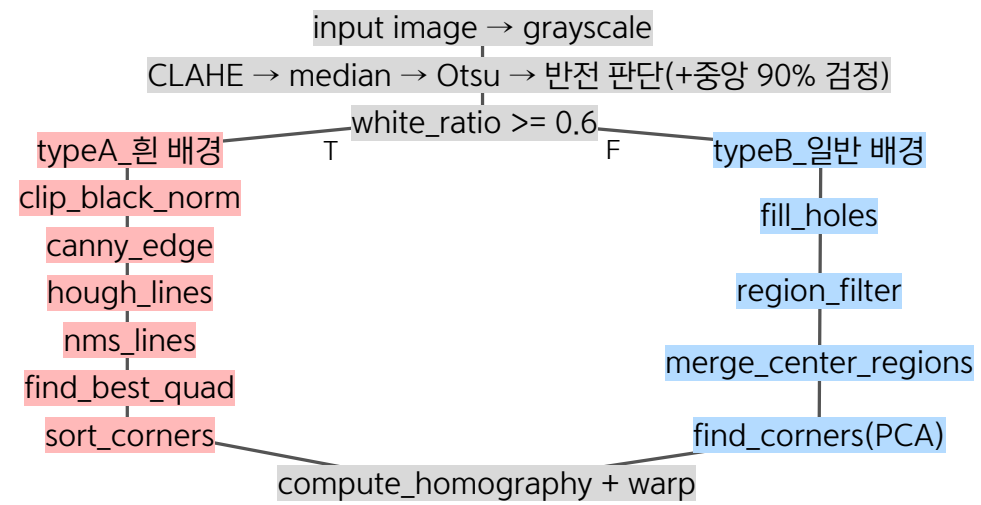
02 typeA/B



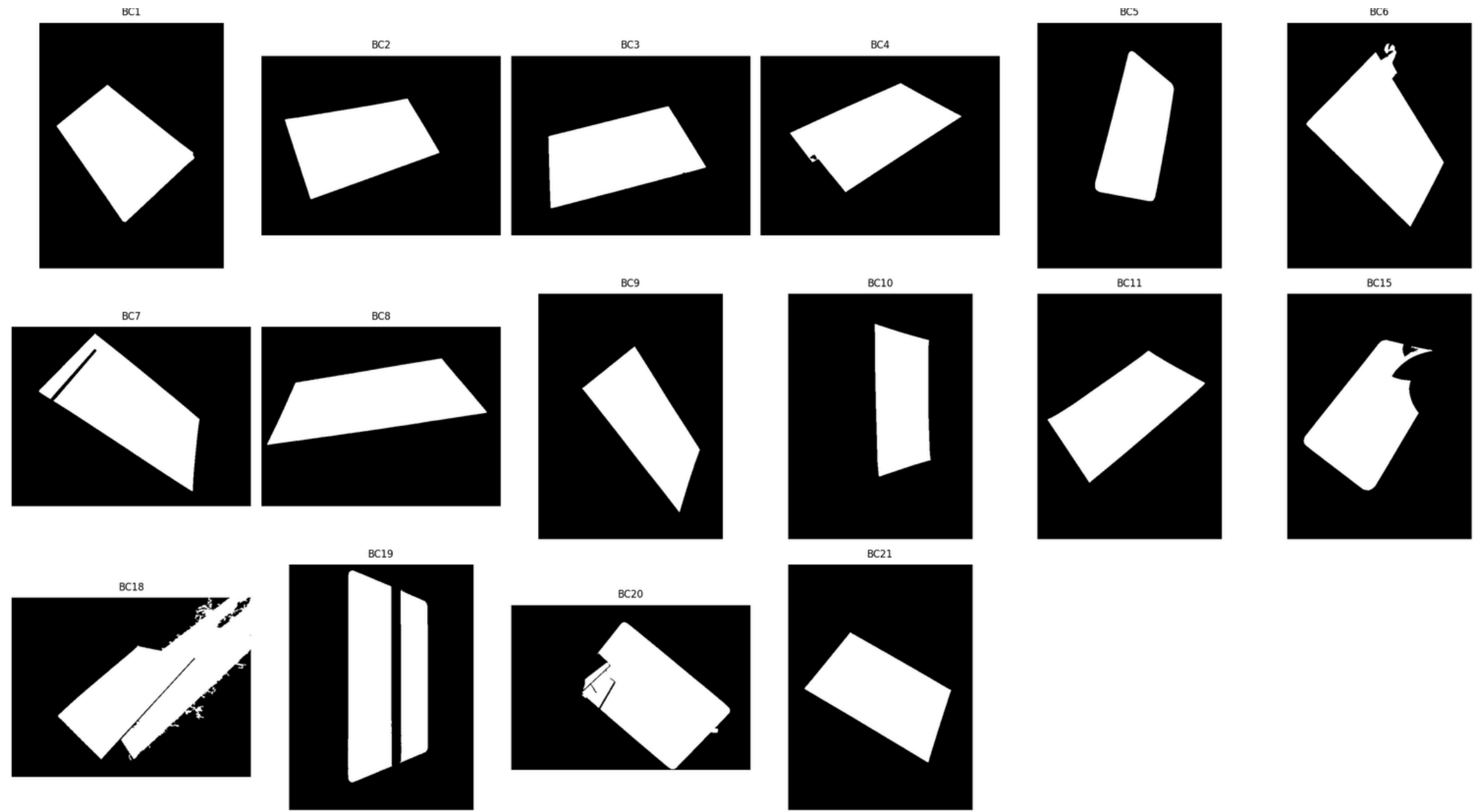
region_filter: 흰 픽셀 전체를 8이웃 BFS로 묶어 component 분리
→ 가장 큰 덩어리만 남기기



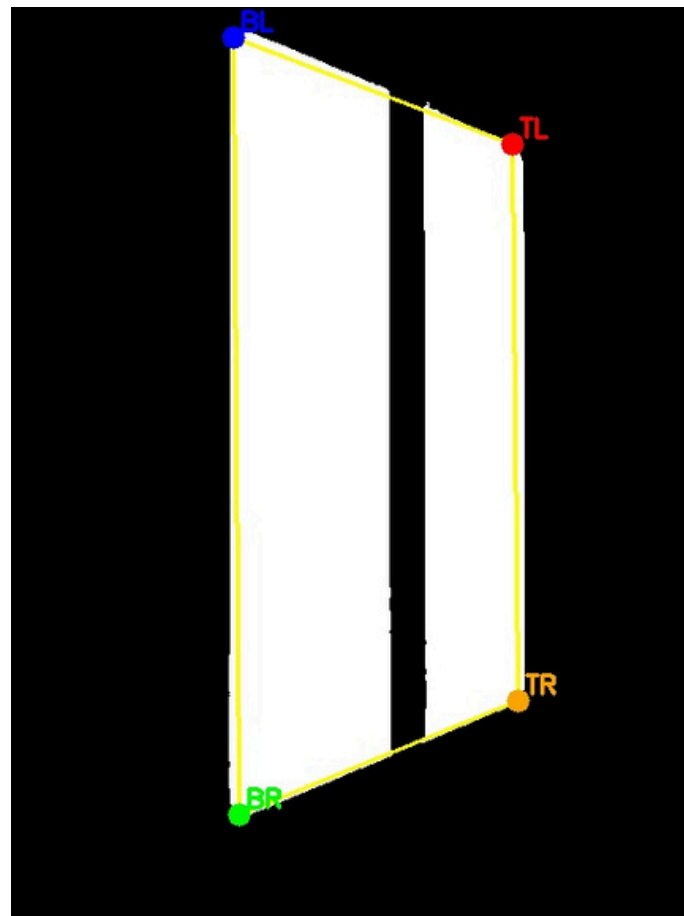
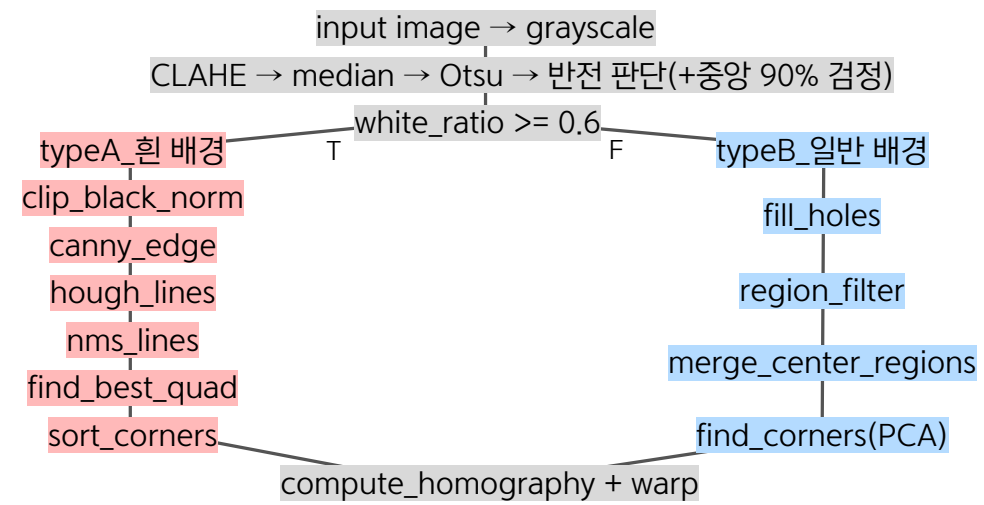
02 typeA/B



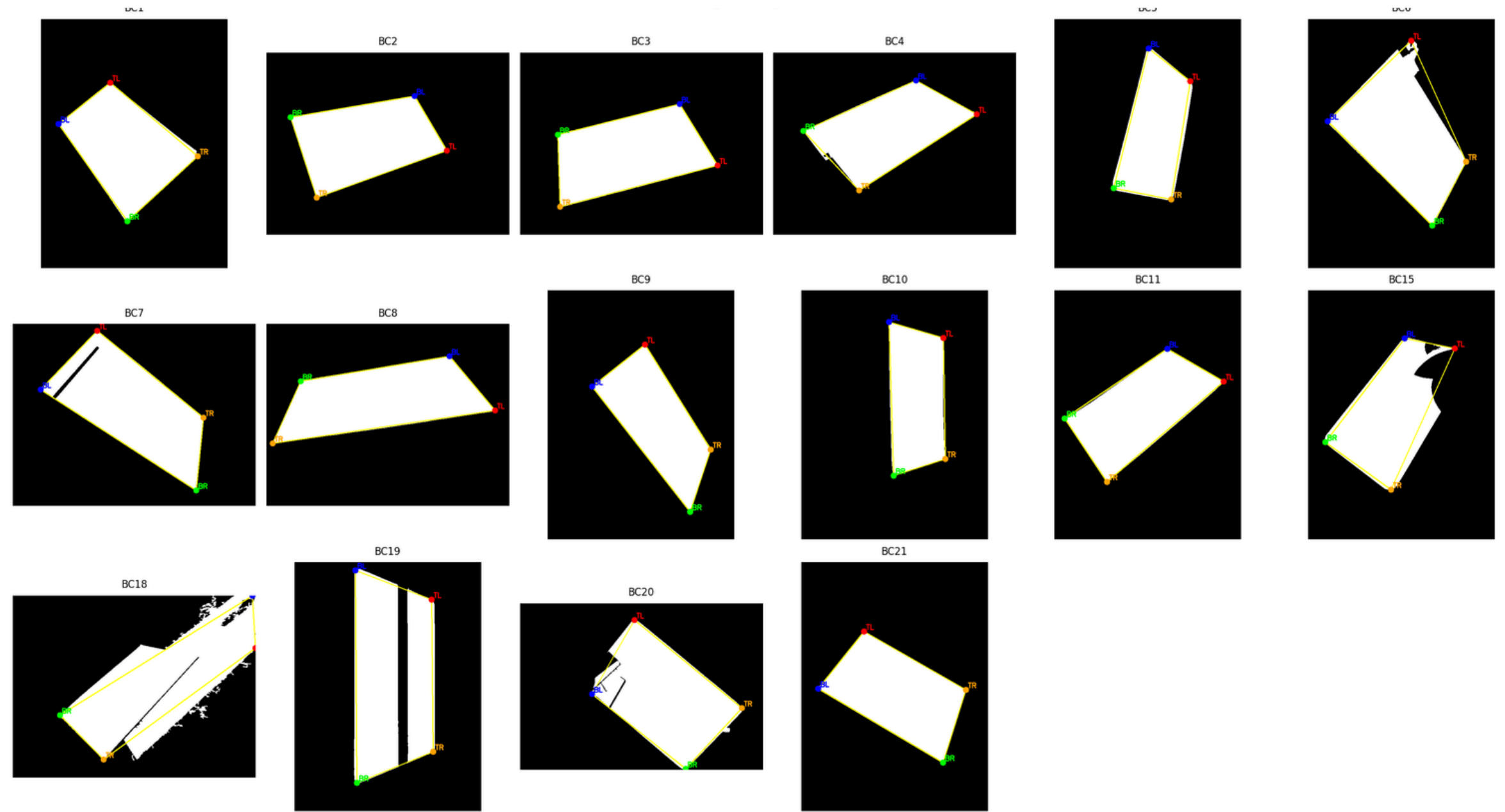
merge: 흑여 구조가 분리된 명함 존재 가능성 생각
→ 중앙 component는 다시 가져와 추가



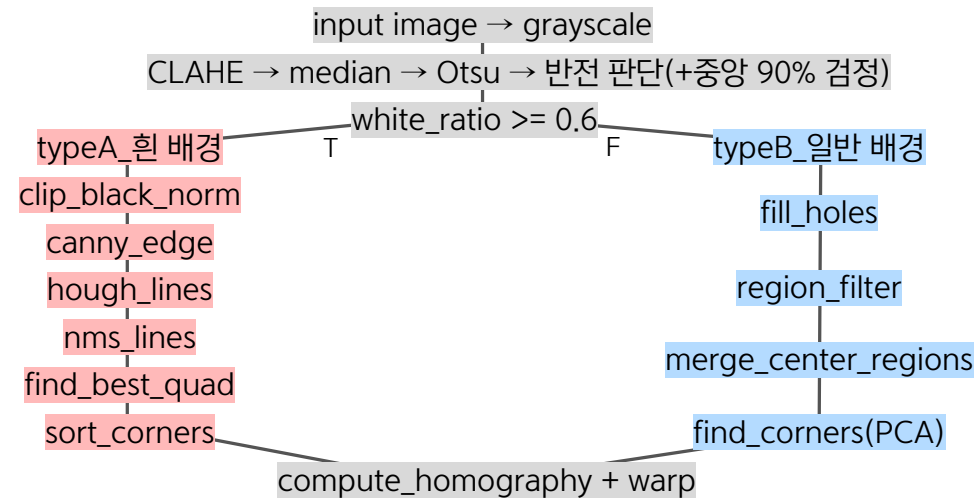
02 typeA/B



find_corners: PCA기반 축 분리 후 회전, 각 극점 = 코너



O2 typeA/B



clip_black_norm:

110 이하 픽셀값 → 0

나머지: 0~255로 norm

밝은 영역 표현력 극대 및 디테일 살

BC12



BC13



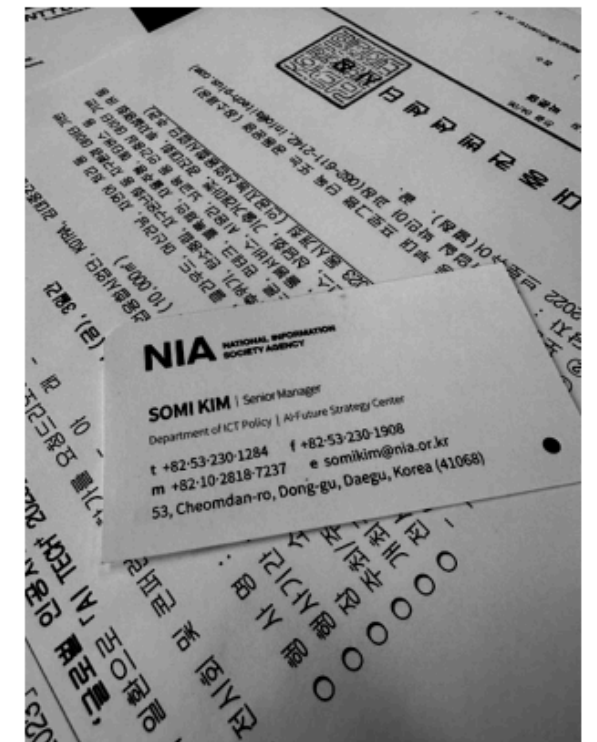
BC14



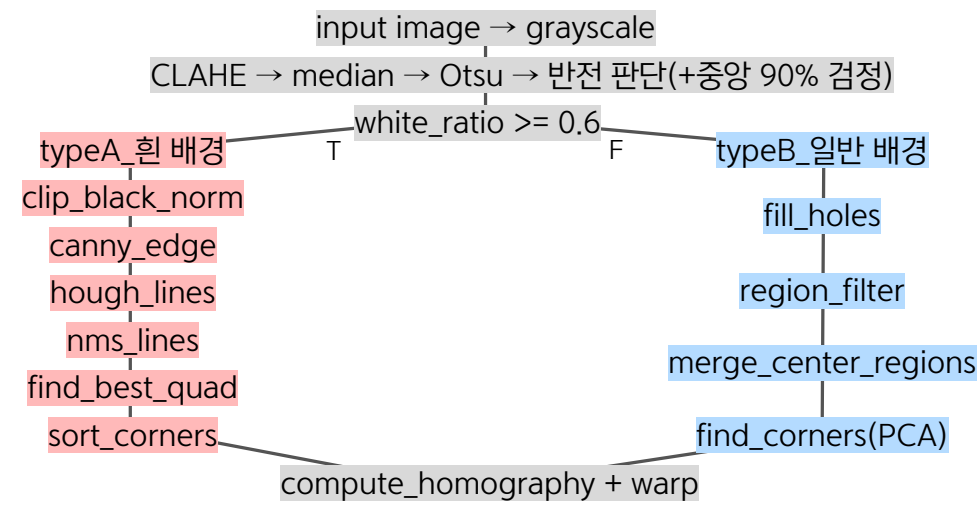
BC16



BC17

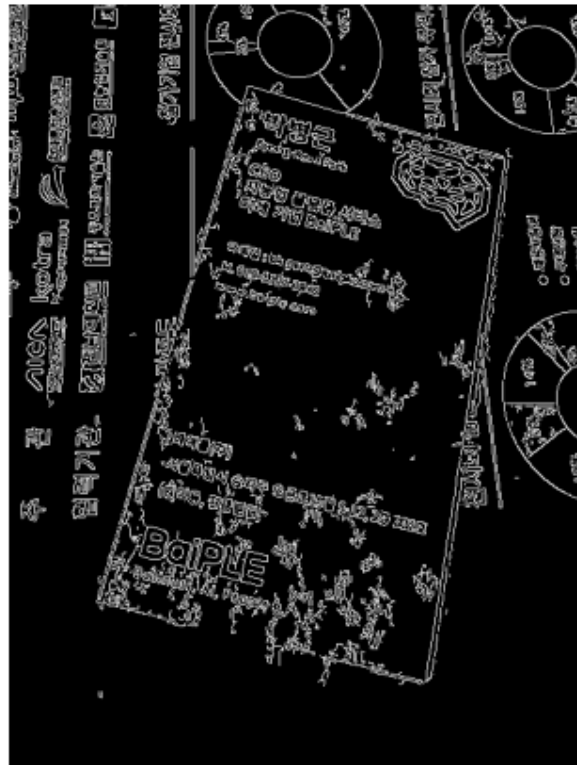


02 typeA/B

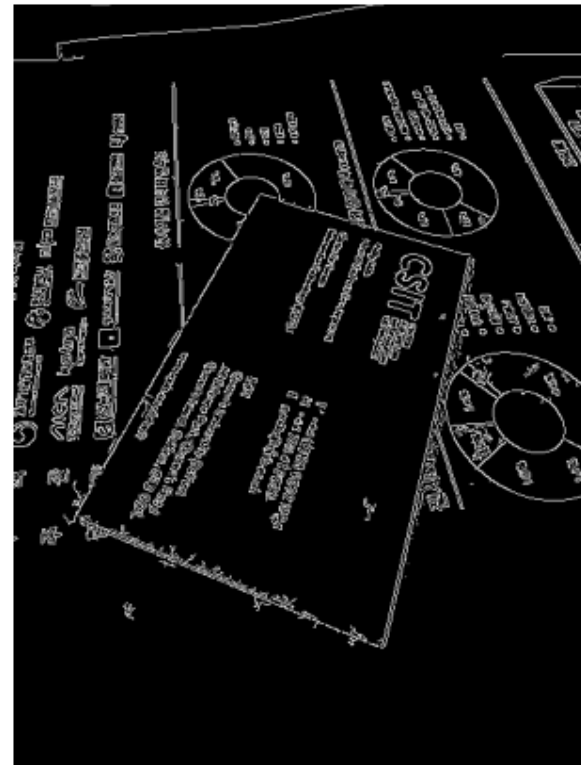


canny edge로 hough 이전 단계 실행

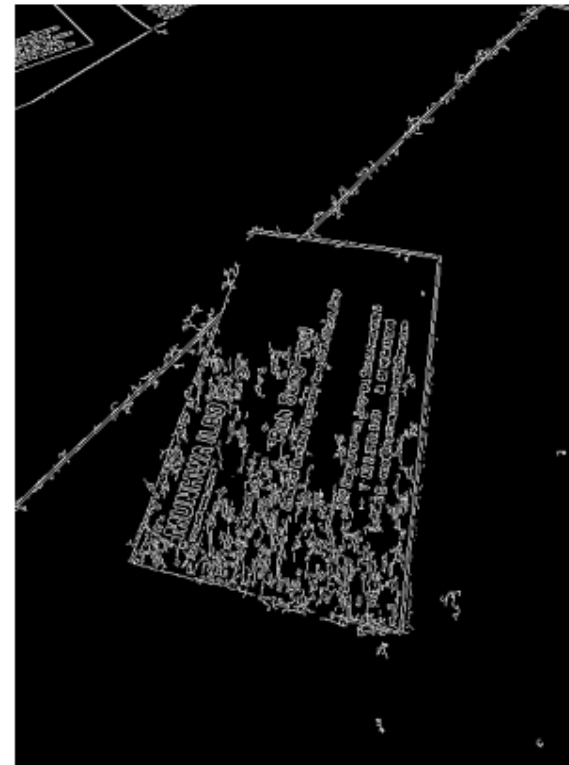
BC12



BC13



BC14



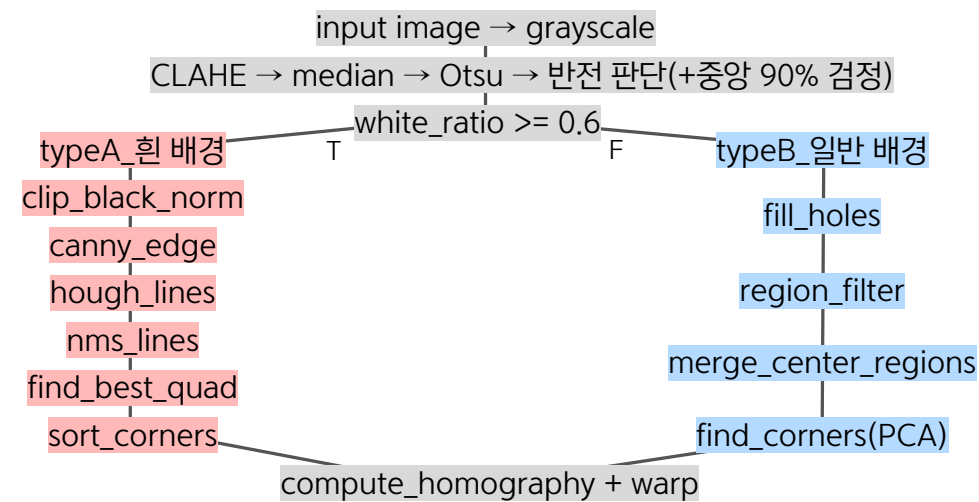
BC16



BC17

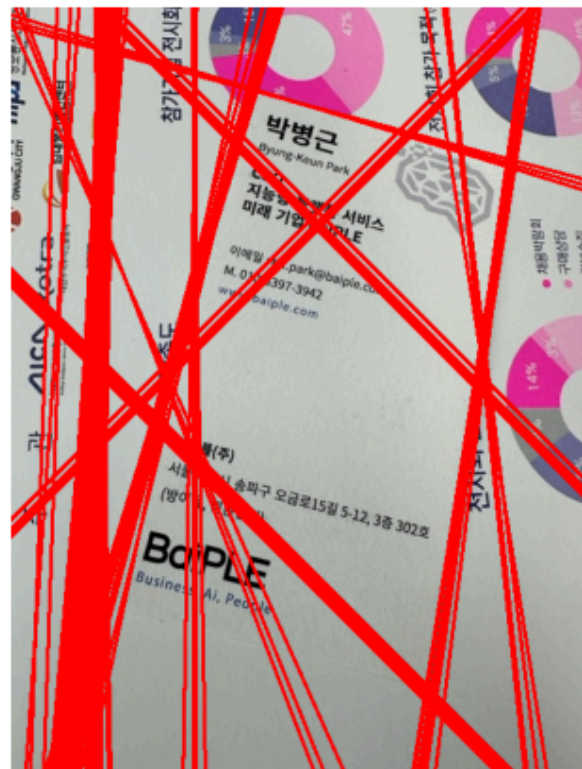


02 typeA/B



hough_lines를 통한 직선 추출
(중복되는 라인이 많아 보임)

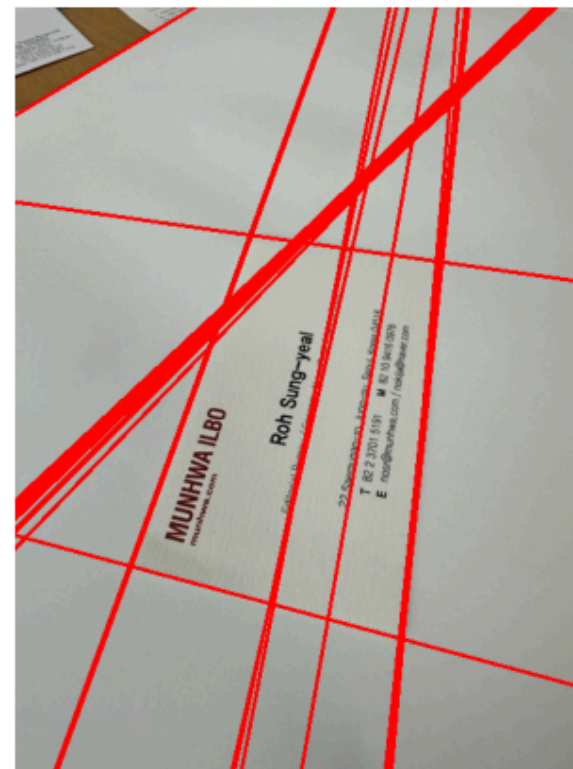
BC12 (99lines)



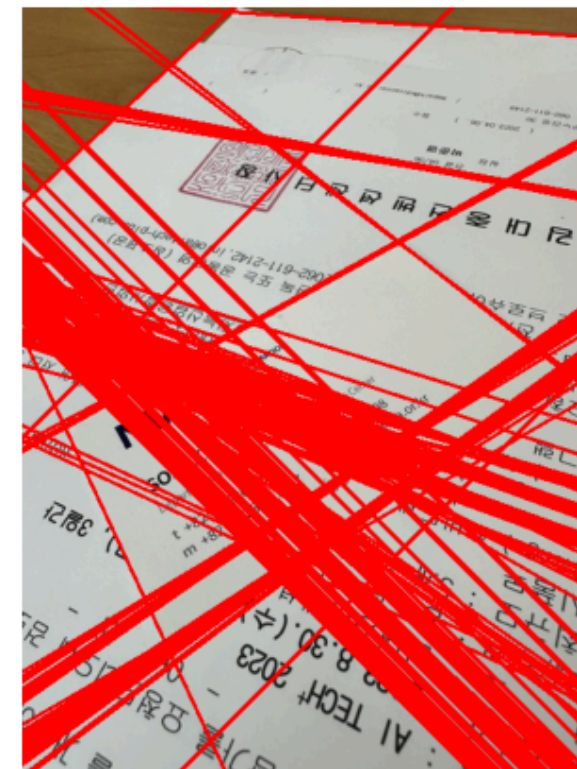
BC13 (29lines)



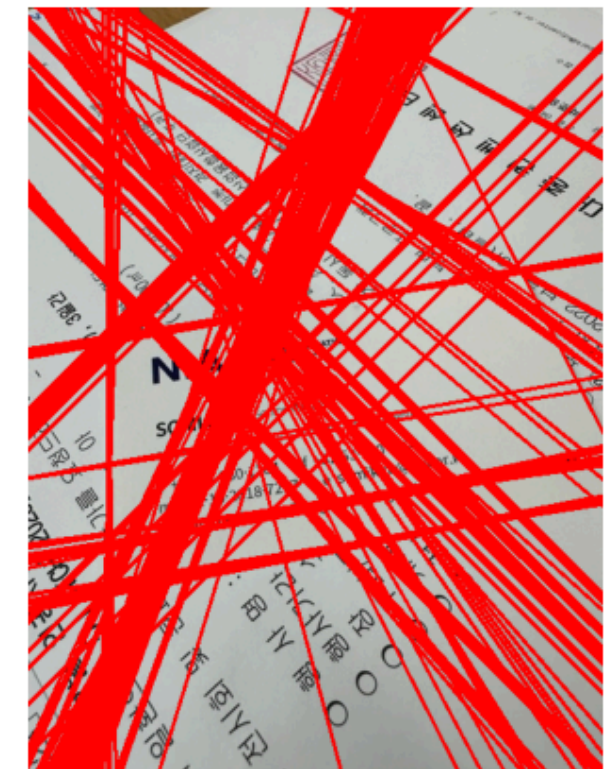
BC14 (28lines)



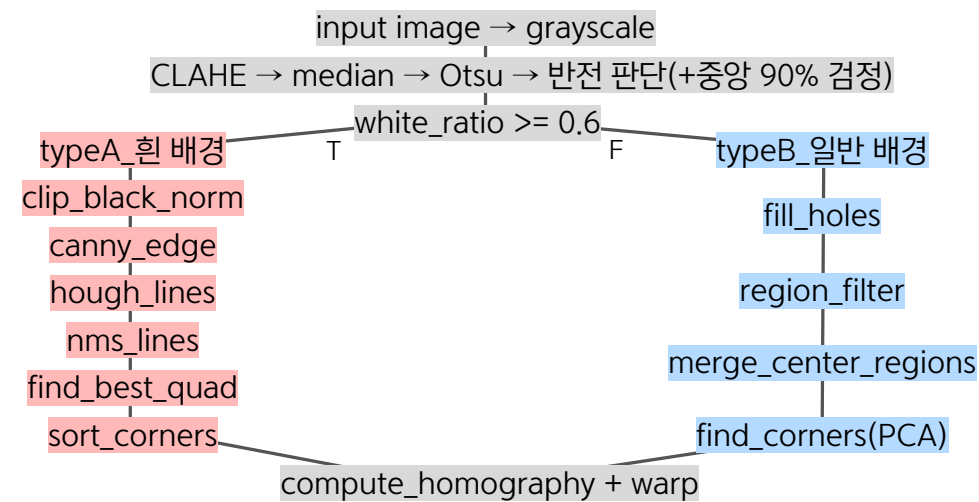
BC16 (169lines)



BC17 (189lines)

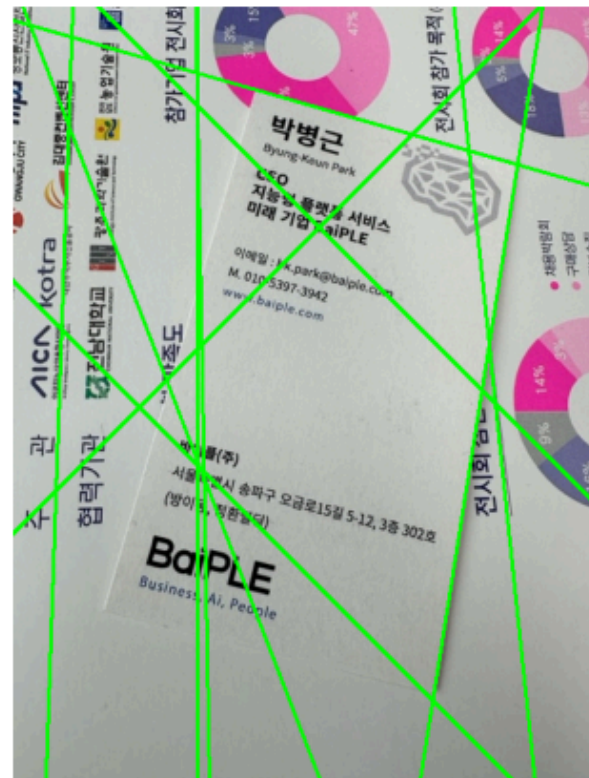


02 typeA/B



nms_lines을 통해 특성을 반영해 라인 억제

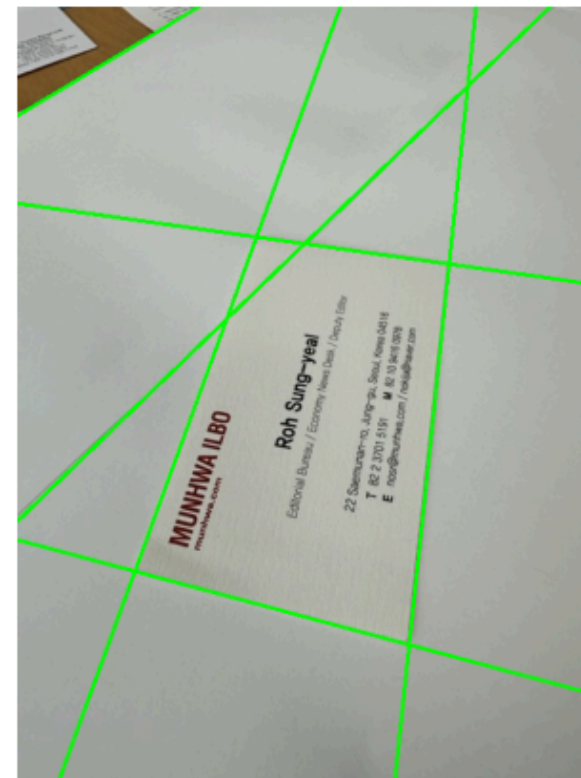
BC12 99→10



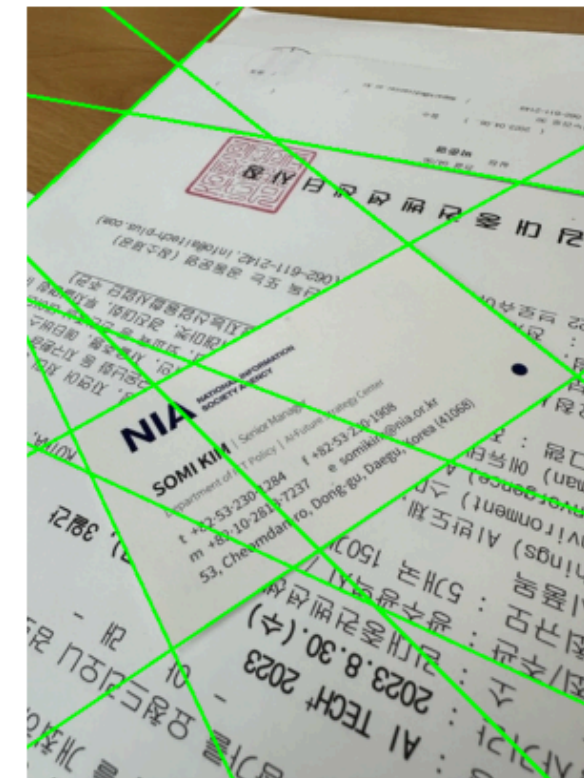
BC13 29→7



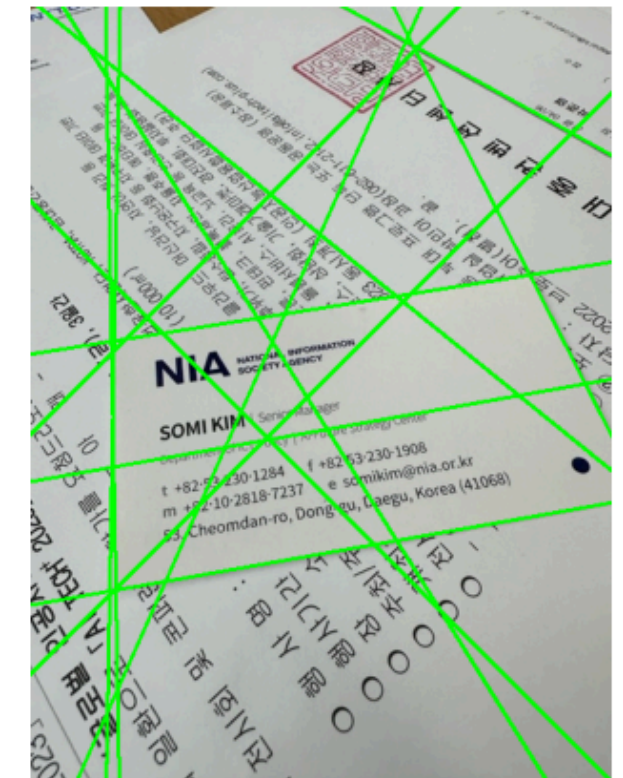
BC14 28→6



BC16 169→9

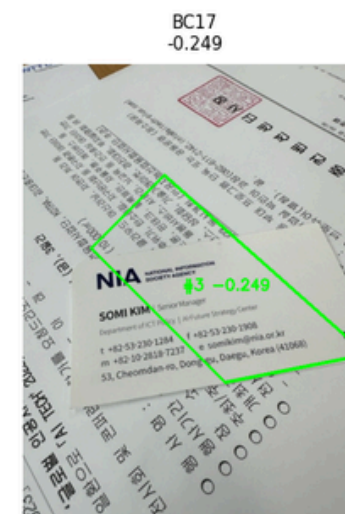
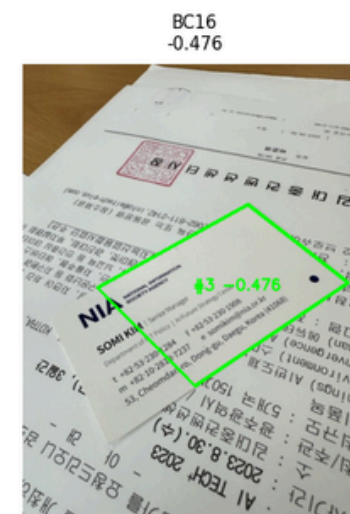
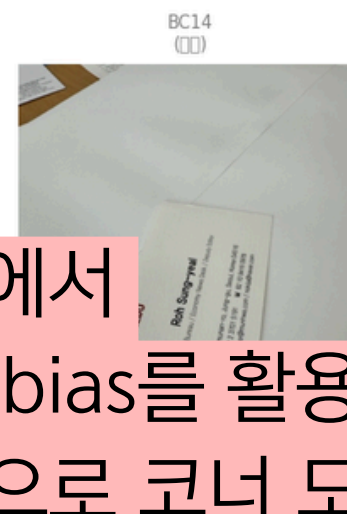
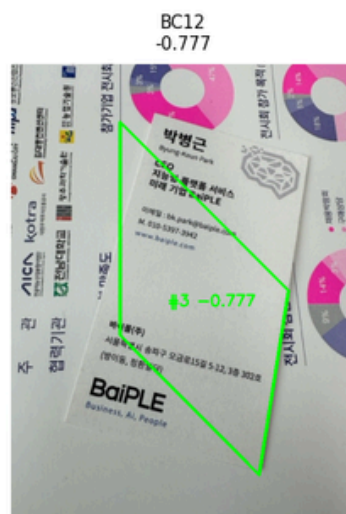
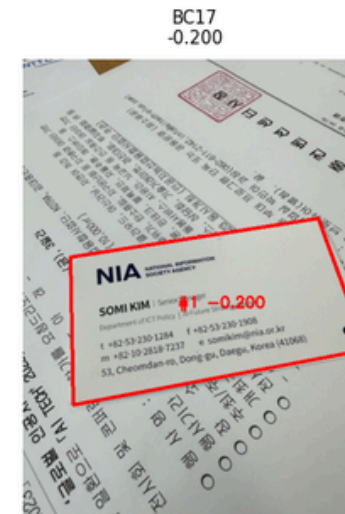
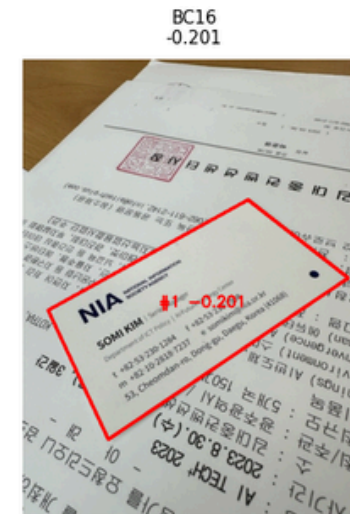
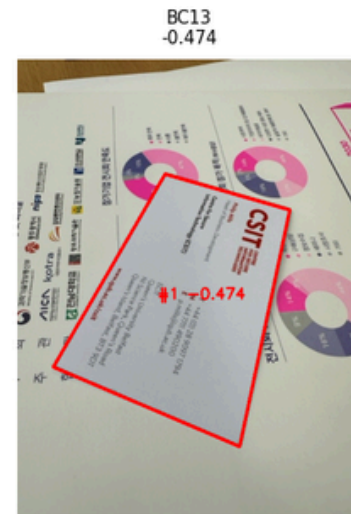


BC17 189→13



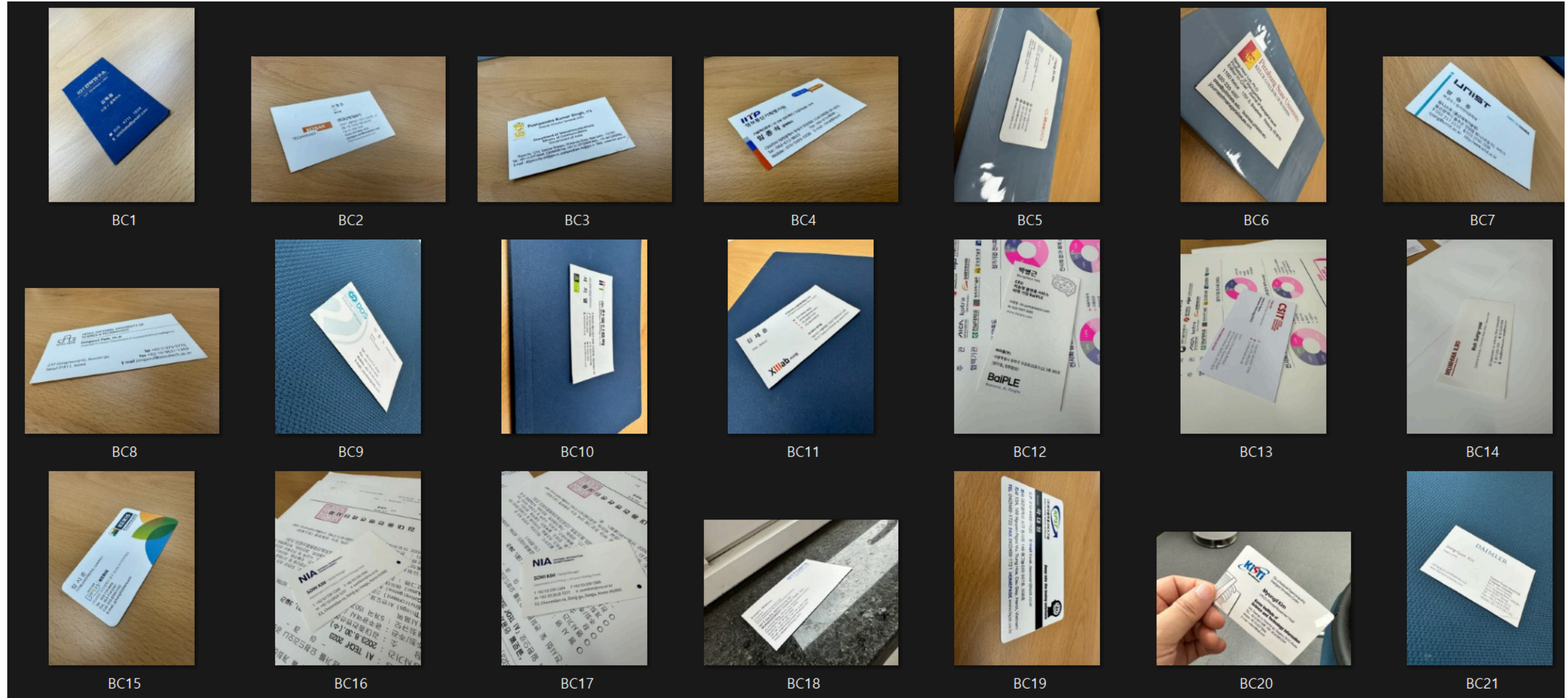
O2 typeA/B

점수 top3



조합 내에서
점수와 inductive bias를 활용해
최적의 직선 조합으로 코너 도출

O3 결과 및 실패 분석



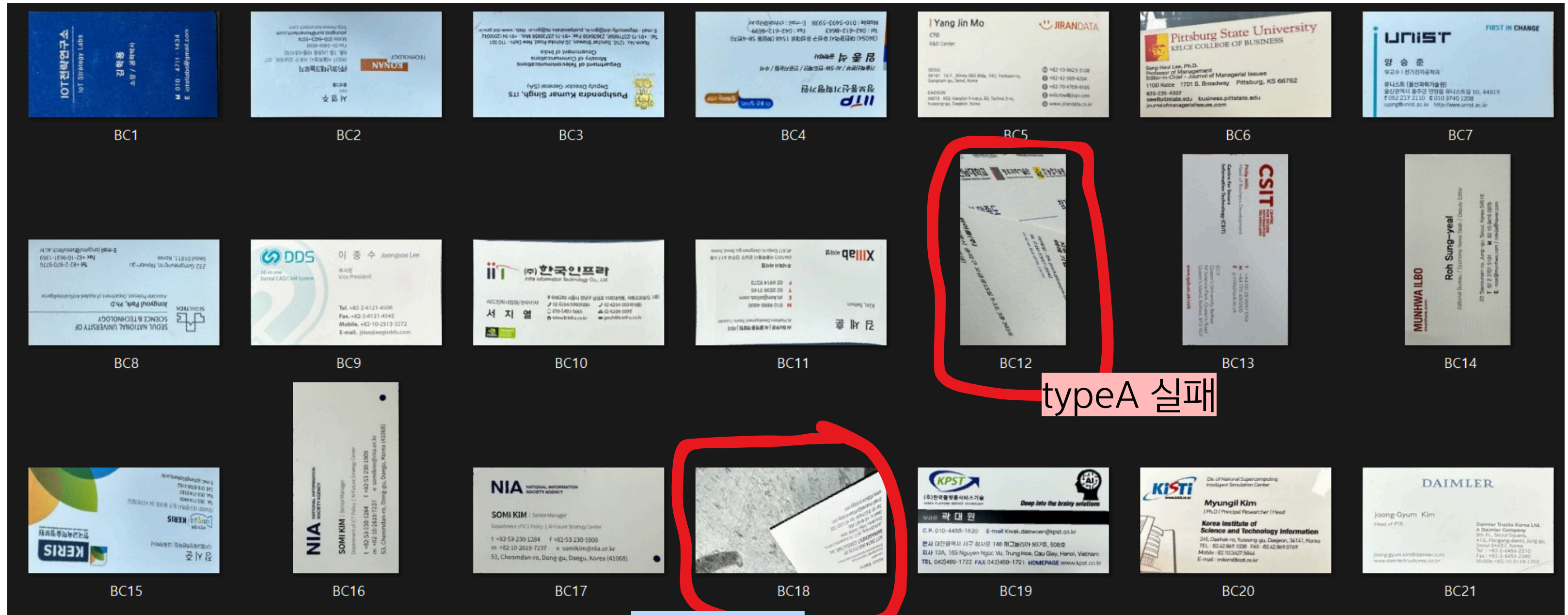
(before)

O3 결과 및 실패 분석



(after)

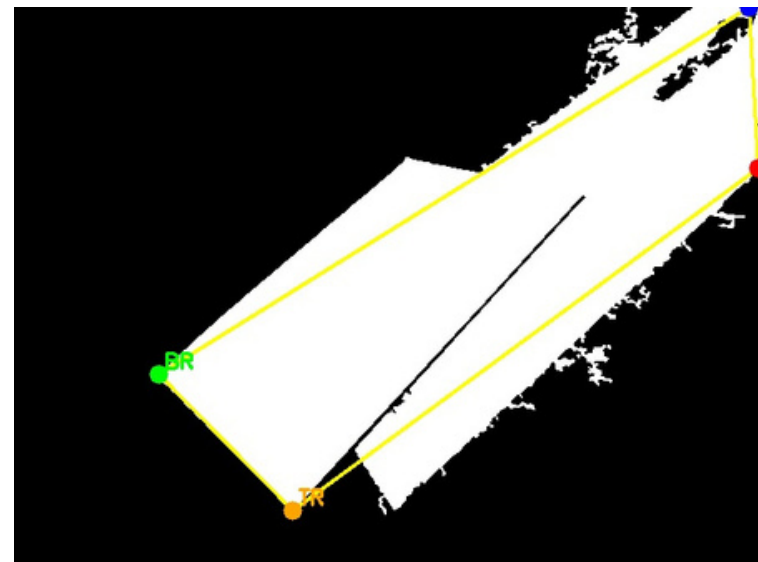
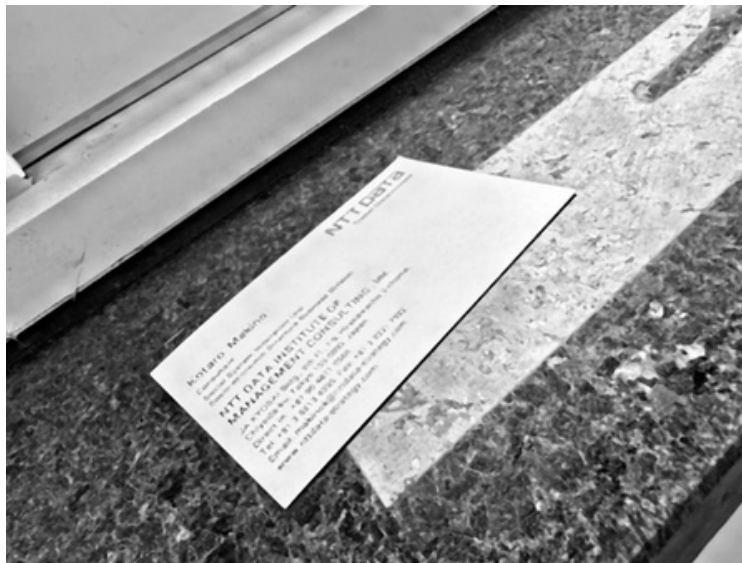
O3 결과 및 실패 분석



typeB 실패
(19/21, 약 90.5% 성공)

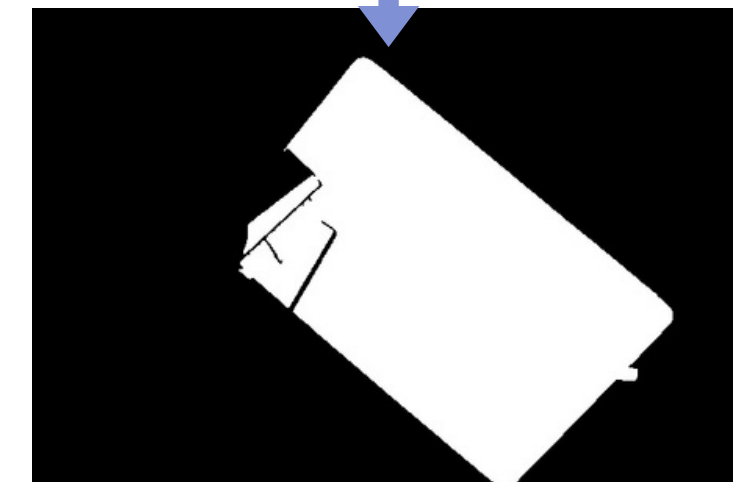
03 결과 및 실패 분석

BC18 typeB 실패 분석



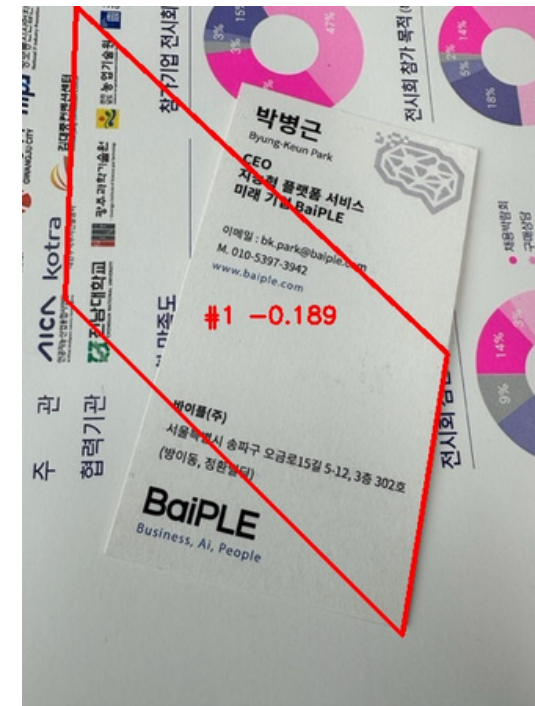
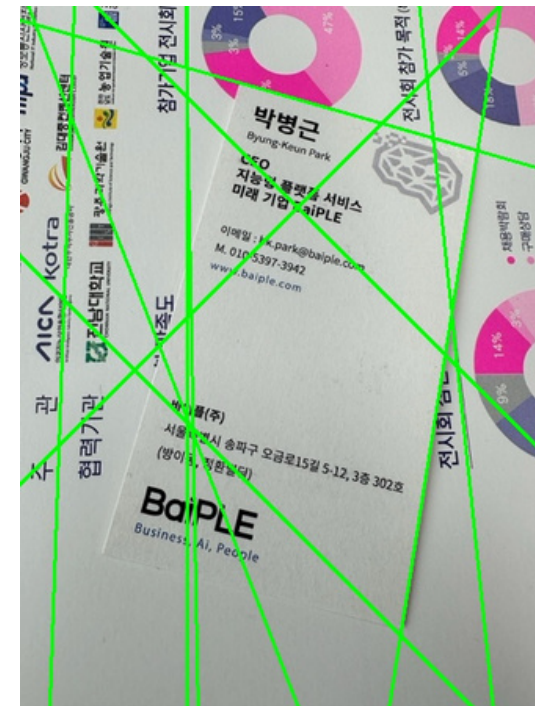
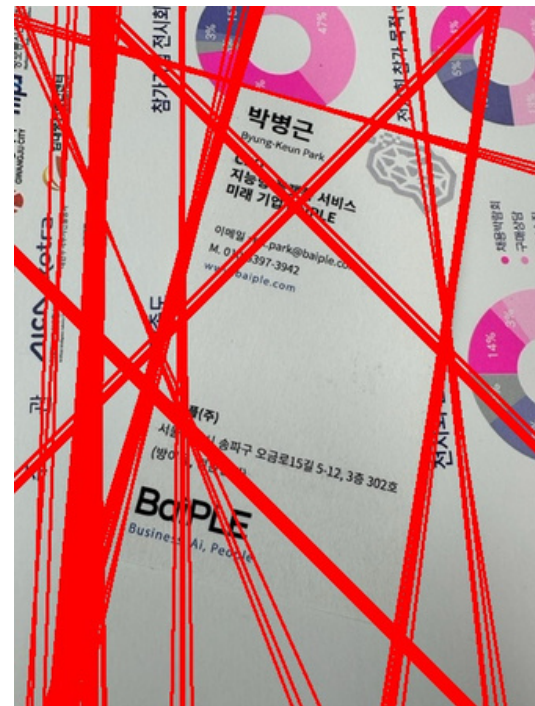
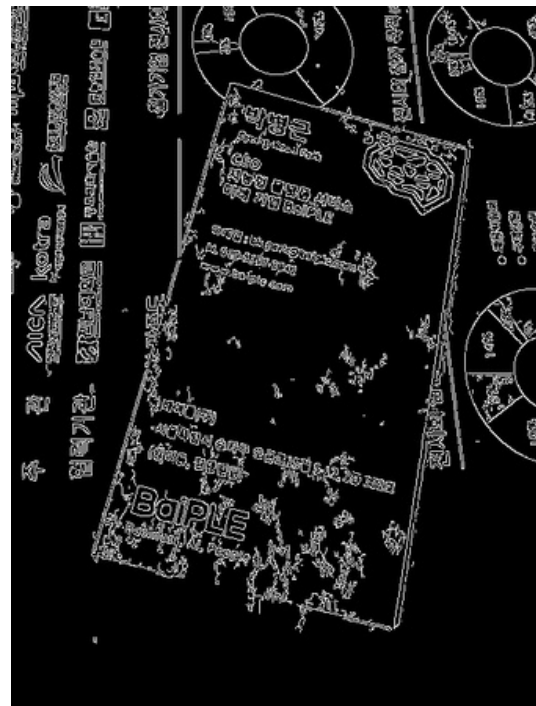
Otsu의 임계값기준 binary
→ 엣지를 구분하지 못하여
덩어리 째로 넘어갔음.
fill_holes를 하더라도,
뒤 반사된 것과 합쳐져
이후 logic 전체에 문제 발생.

cf) BC20의 코너 부분에서,
gaussian에서 median으로 변경 후
손가락이 분리되어 좋은 결과를
얻을 수 있었음.



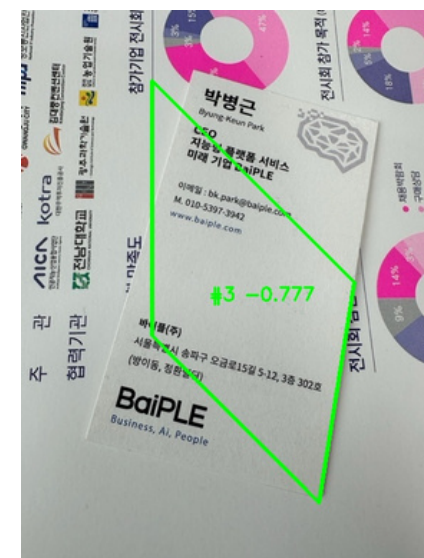
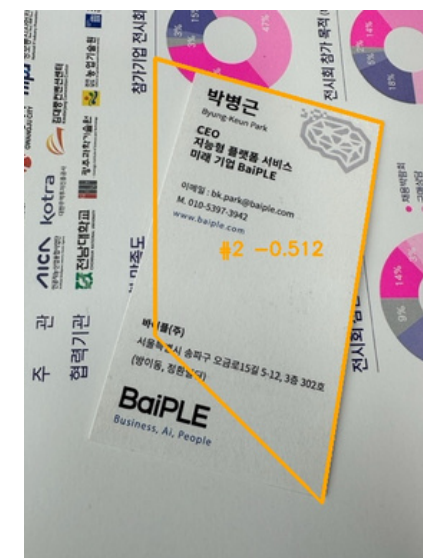
03 결과 및 실패 분석

BC12 typeA 실패 분석



canny → hough 과정에서
threshold 120개의 점이어야 line 검출

그러나, 해당 명함 하단의 경계는 직선이 생기지 않았음.
100으로 낮출 경우 검출이 되나,
직선의 개수가 기하급수적으로 커져 이후 logic 적용이 힘들



THANK YOU

감사합니다